

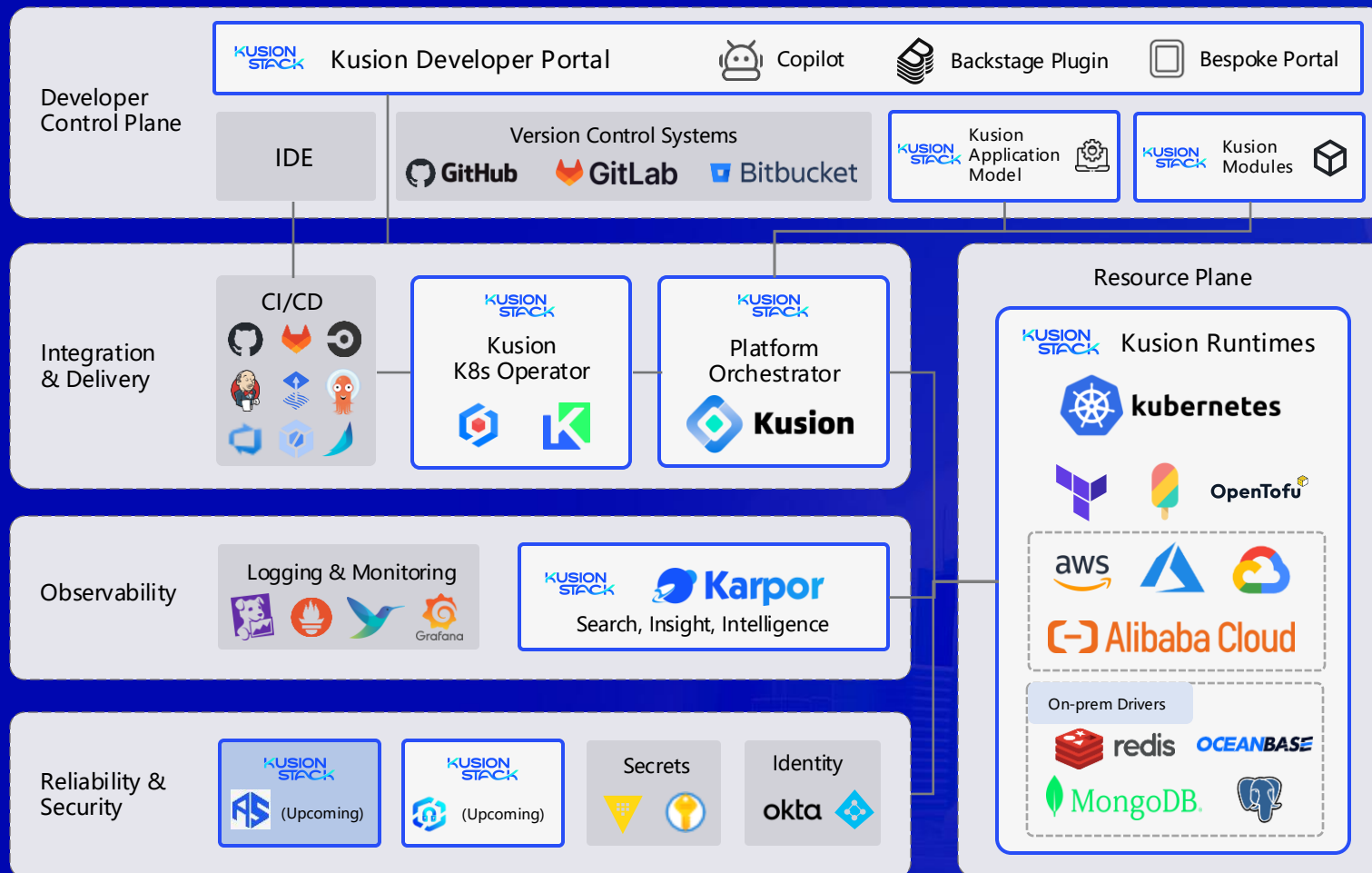
Karpor: Navigating the Multi-Cluster Maze with Ease Using a Kubernetes Explorer

Forest Chen | KusionStack Maintainer | Ant Group

2025.08.19

KusionStack: Platform Engineering Stack

KusionStack for Internal Developer Platforms



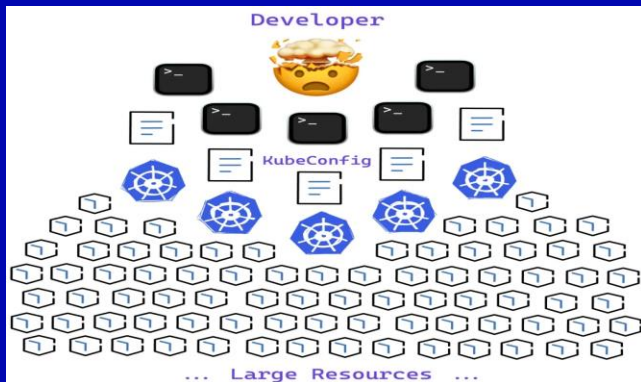
What is KusionStack?

- Platform Engineering Building Blocks
- Loosely-coupled set of tools
- Voted into CNCF Sandbox in 2024

What does it include?

- Kusion: Intent-Driven Platform Orchestrator
- Kuperator: Set of Kubernetes controller extensions
- Karpenter: Multi-cluster Kubernetes Dashboard, focused on search & insight
- ...

Background in Multi-cluster management

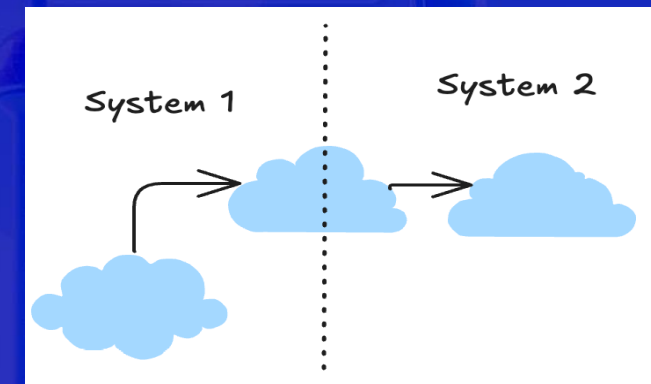


$O(M*N)$ complexity to manage M clusters with N users, systems or platforms, especially in multi-cloud/multi-region settings

The complexity of Kubernetes itself including its upstream and downstream systems grows over time



The need to improve effectiveness of infrastructure and platform engineers by embracing new technologies like LLMs



What do our users need?

【Quickly Locate Resources】 How do I find a specific (set of) resource(s) across tens or hundreds of clusters? Where are my applications located? What resources do they include?

【Resource Topologies】 How does a group of logically-relevant resources relate to each other?

【Logical View】 How to map Kubernetes resources to business-related domain models such as applications or environments?

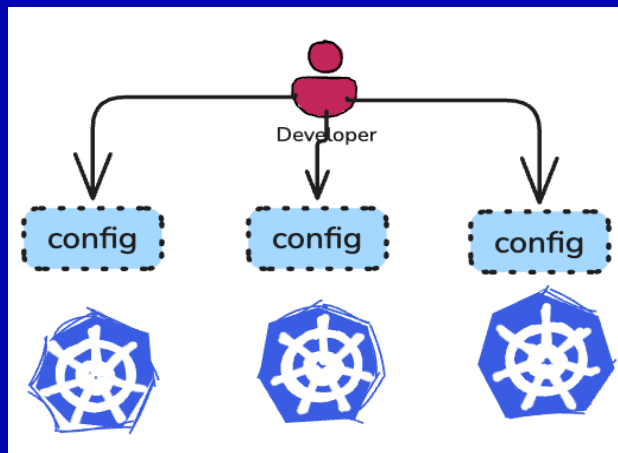
【Security and Compliance】 Are my clusters/resources safe? Are there any unflagged risks, vulnerabilities or attack vectors?

【Troubleshooting】 Did someone alter a particular resource? Is there a visualized timeline of changes?

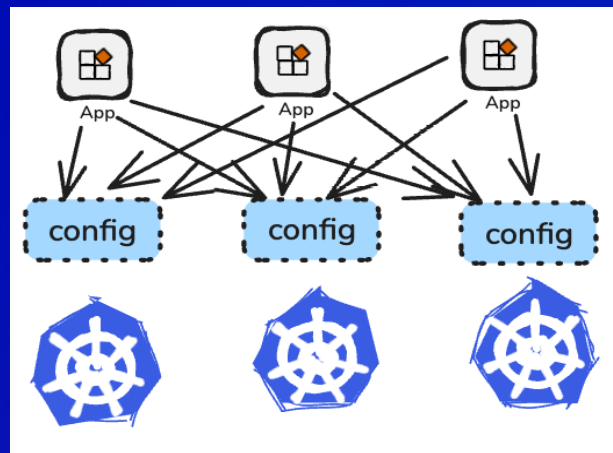
.....



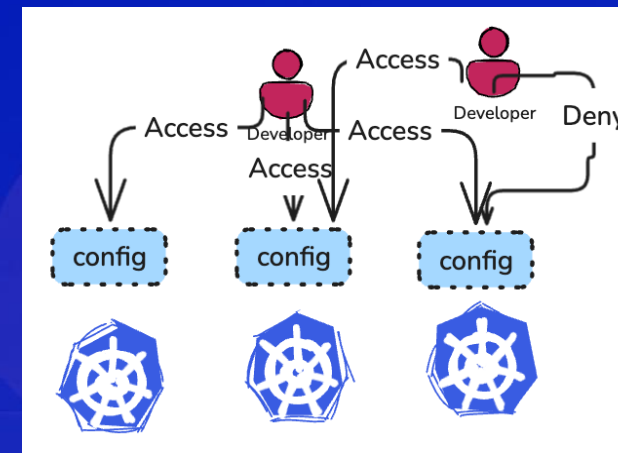
Pain Points



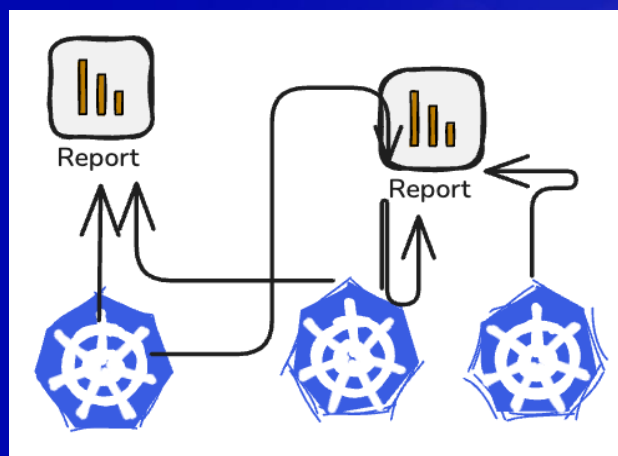
User Credential Management



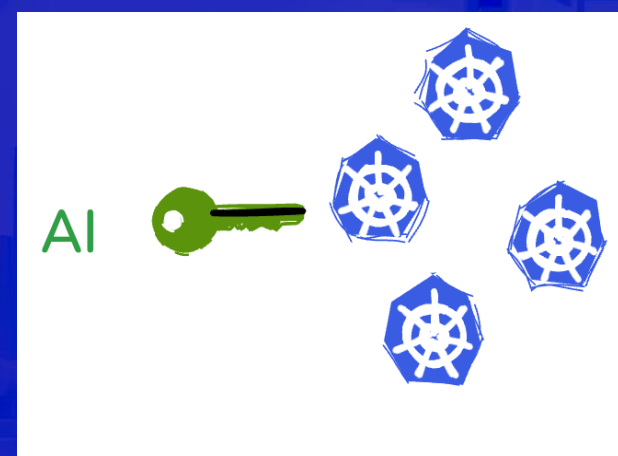
Complex System Integrations



Convolutd Permissions



Hard to Extract Helpful Insights



Need to improve efficiency

? Problems with...

Credential
Management

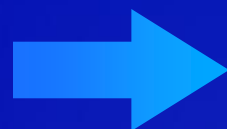
System
Integrations

Complex
Permissions

Cross-cluster
Search

Insight
Extraction

Empower
with AI



★ Ideas

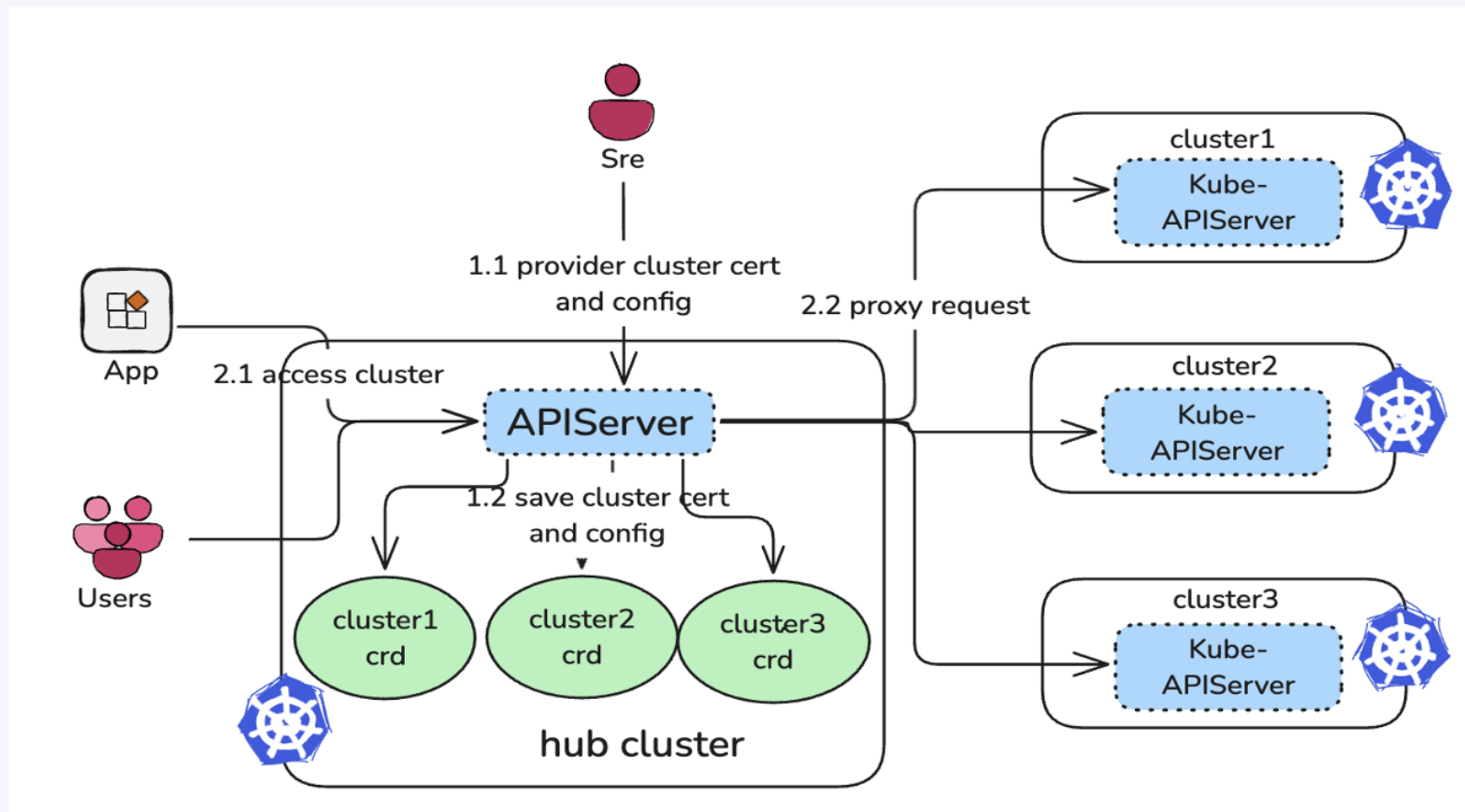
Multi-cluster Proxy, Single Entrance
with Unified Permission Management

Real-time HA Data Synchronization
into storage, w/ Transformations

Search and Insights with single,
aggregated and custom views

AI for Kubernetes. Analyze data,
produce reports and search with NL

Multi-cluster Gateway Proxy



Register w/ K8s certificate

High performance

RBAC w/ Impersonation supported

IAM for Access Control

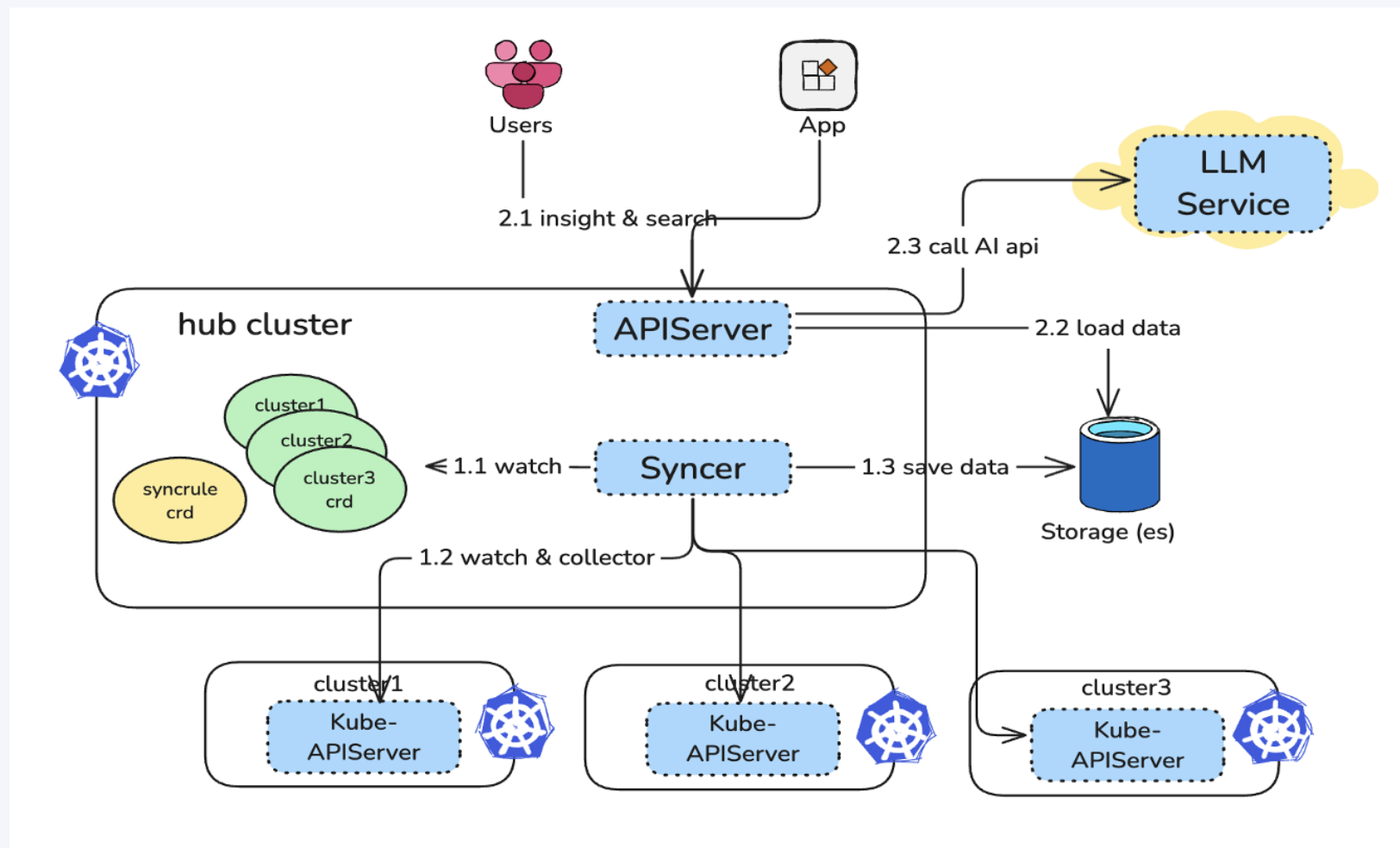
K8s-native Requests

Registration: SREs by providing the KubeConfig. The gateway stores them as Cluster CR.

Usage: Direct multi-cluster traffic by sending them to the /proxy endpoint on the gateway.

Example: GET /apis/cluster.addon.open-cluster-management.io/clusterextensions/{cluster1}/proxy/api/v1/pods

Data Synchronization and Insights



Search with...



KEY



SQL



NLP

Insights on...



Compliance

Reports



Event Timelines

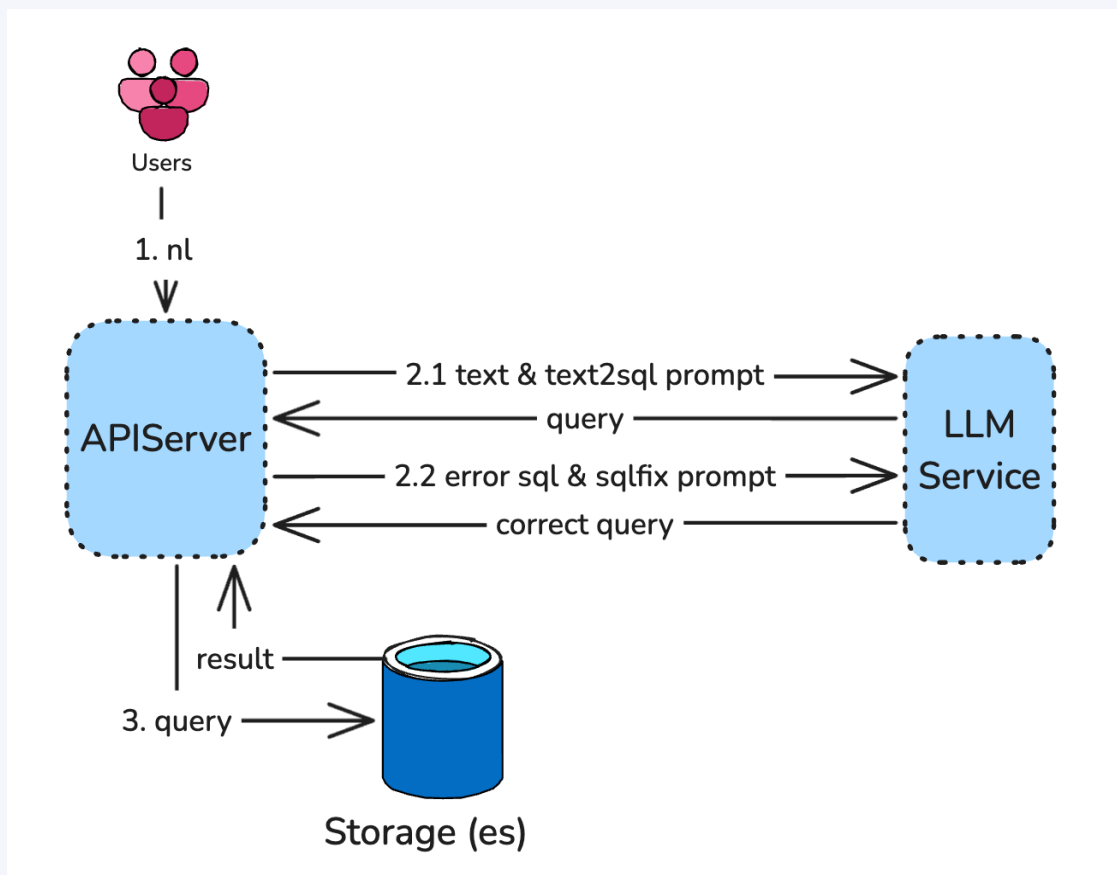


Resource

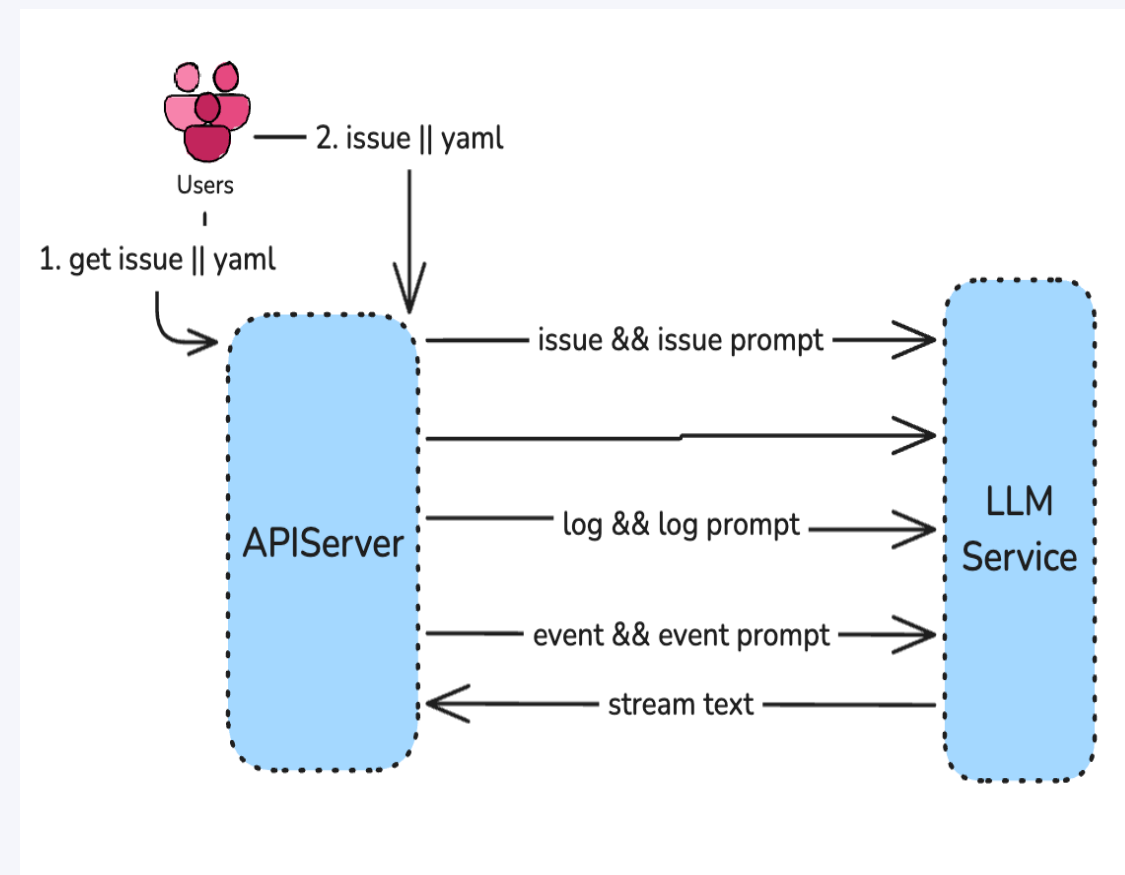
Topologies

Sync: Data Synchronization based on SyncRule and Cluster CRs

Consume: Apps and Users perform search and get insights via the API Server and Dashboard

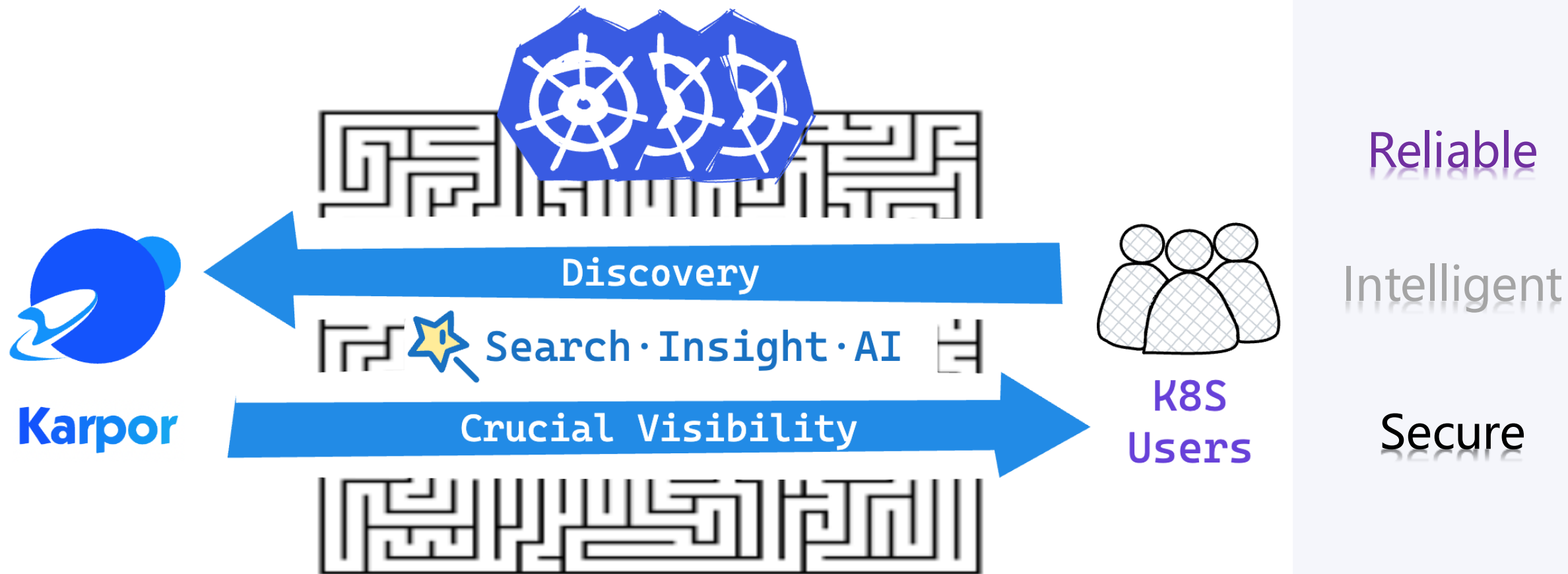


Convert Input to search query
Search with NL



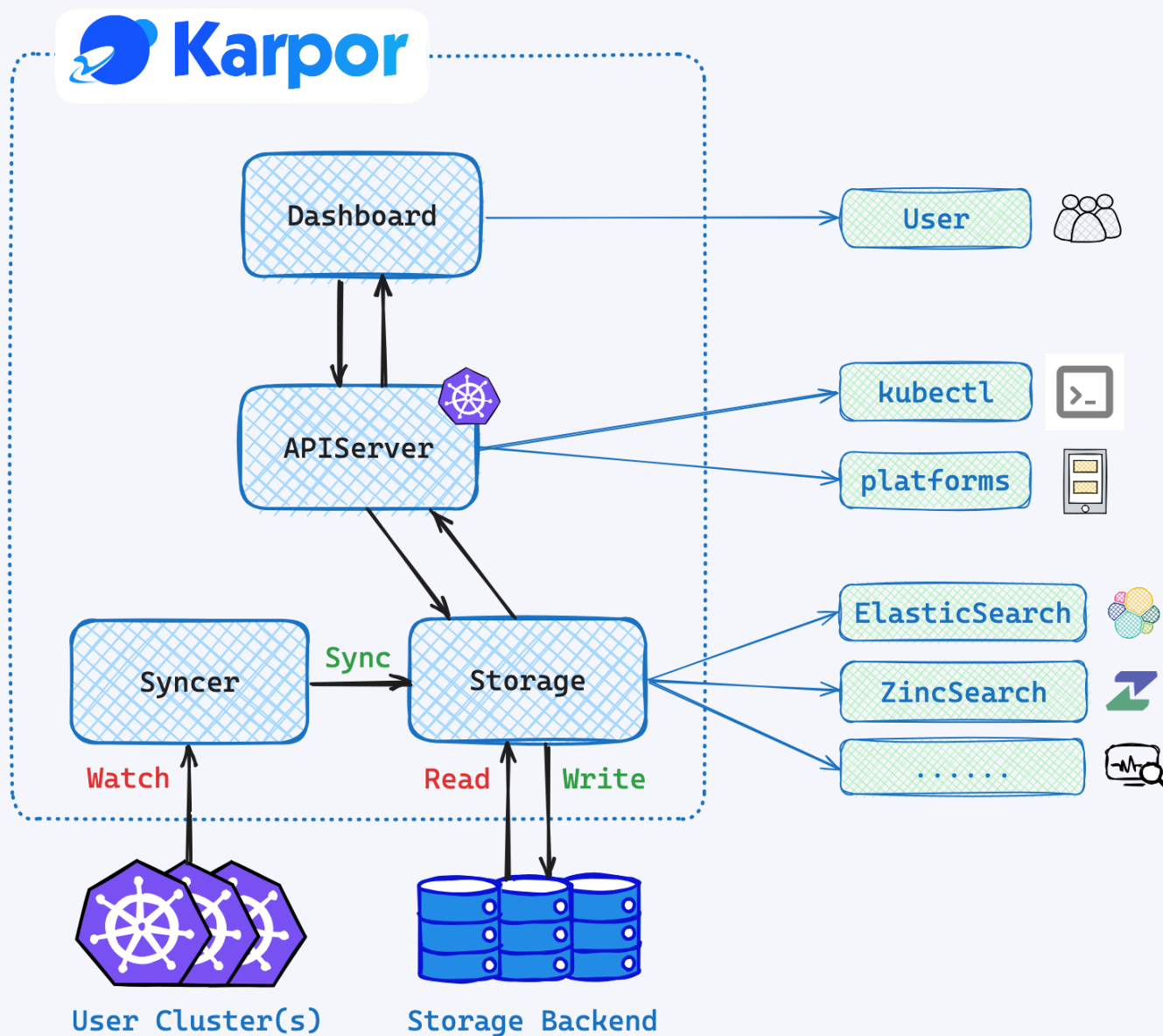
Use LLM to Analyze YAMLs/logs/events
Identify Issues early and generate fixes

Karpor is all you need



Karpor is all you need

Architecture Overview

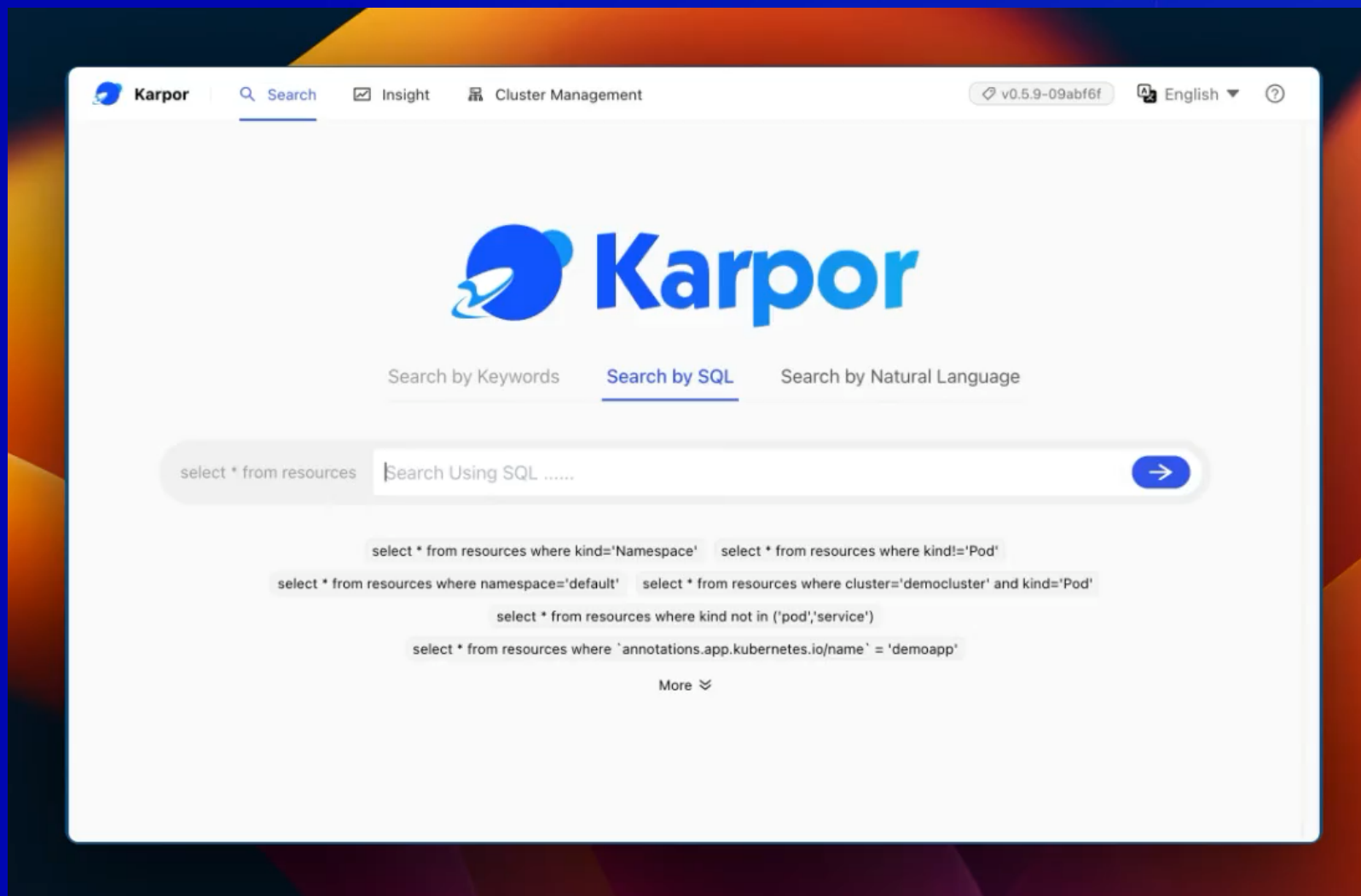


Components:

- Dashboard
- Aggregated APIServer
- Syncer
- Storage Adapter

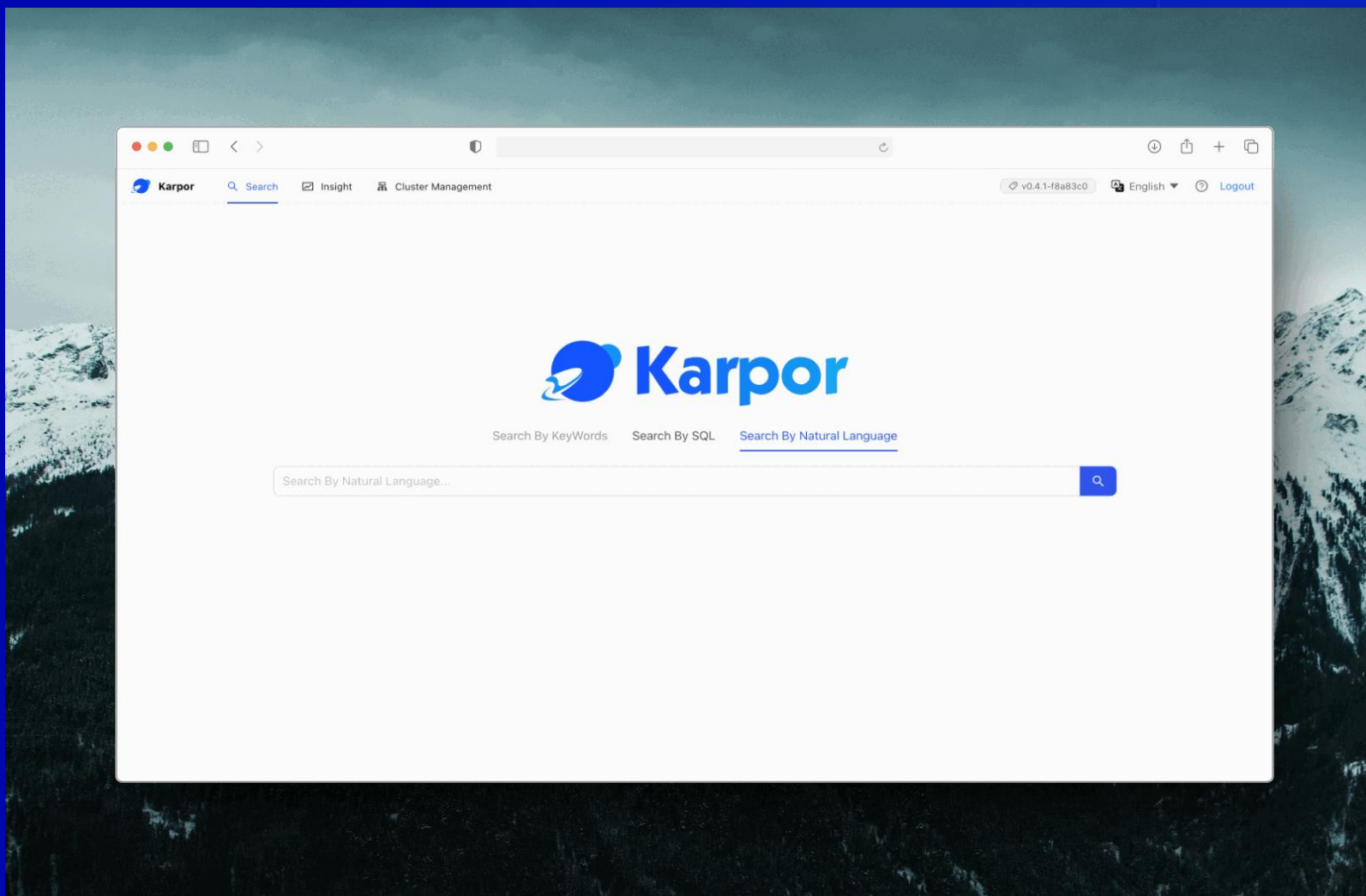
Search-focused: Kubernetes Dashboard

Search with Keywords or SQL, optimized for Kubernetes



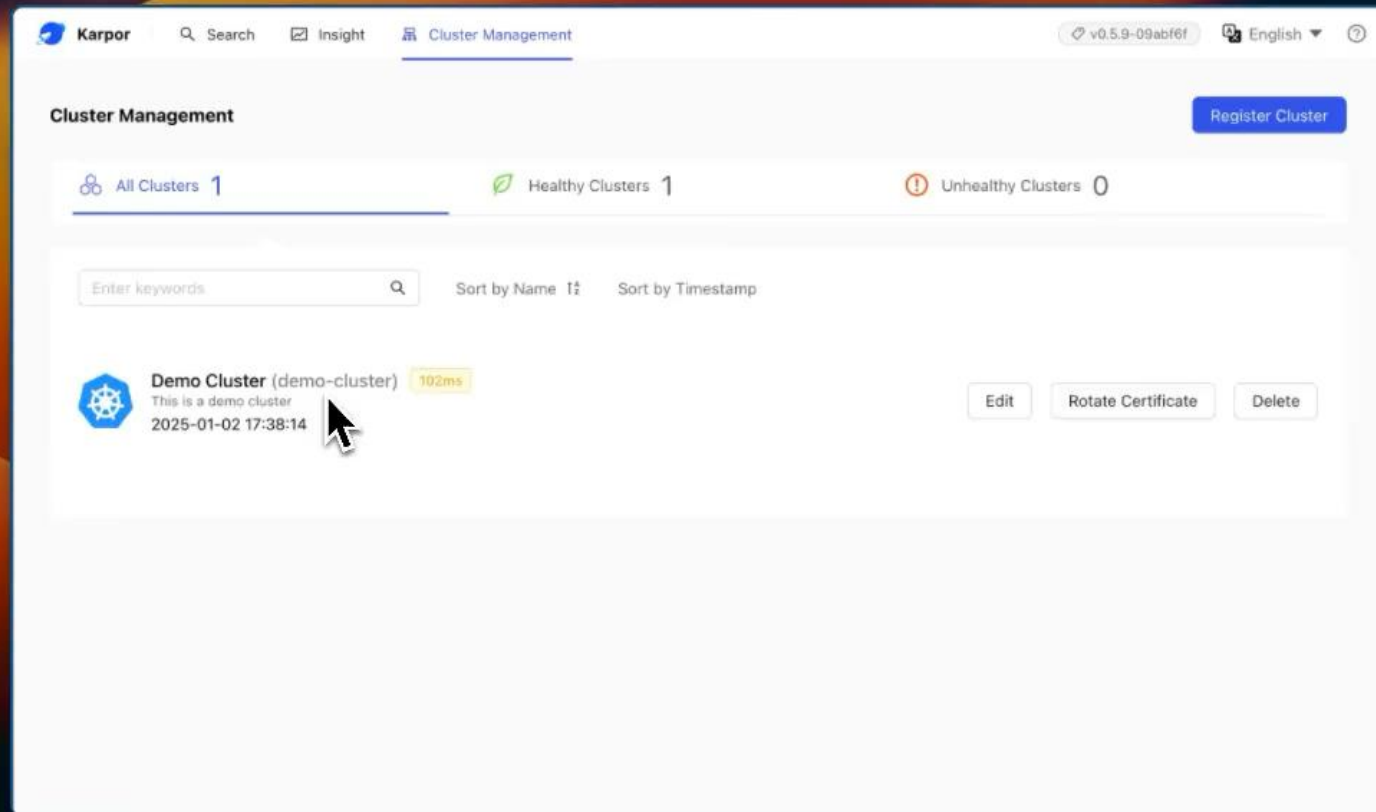
Search-focused: Kubernetes Dashboard

More Intuitively Search with Natural Language



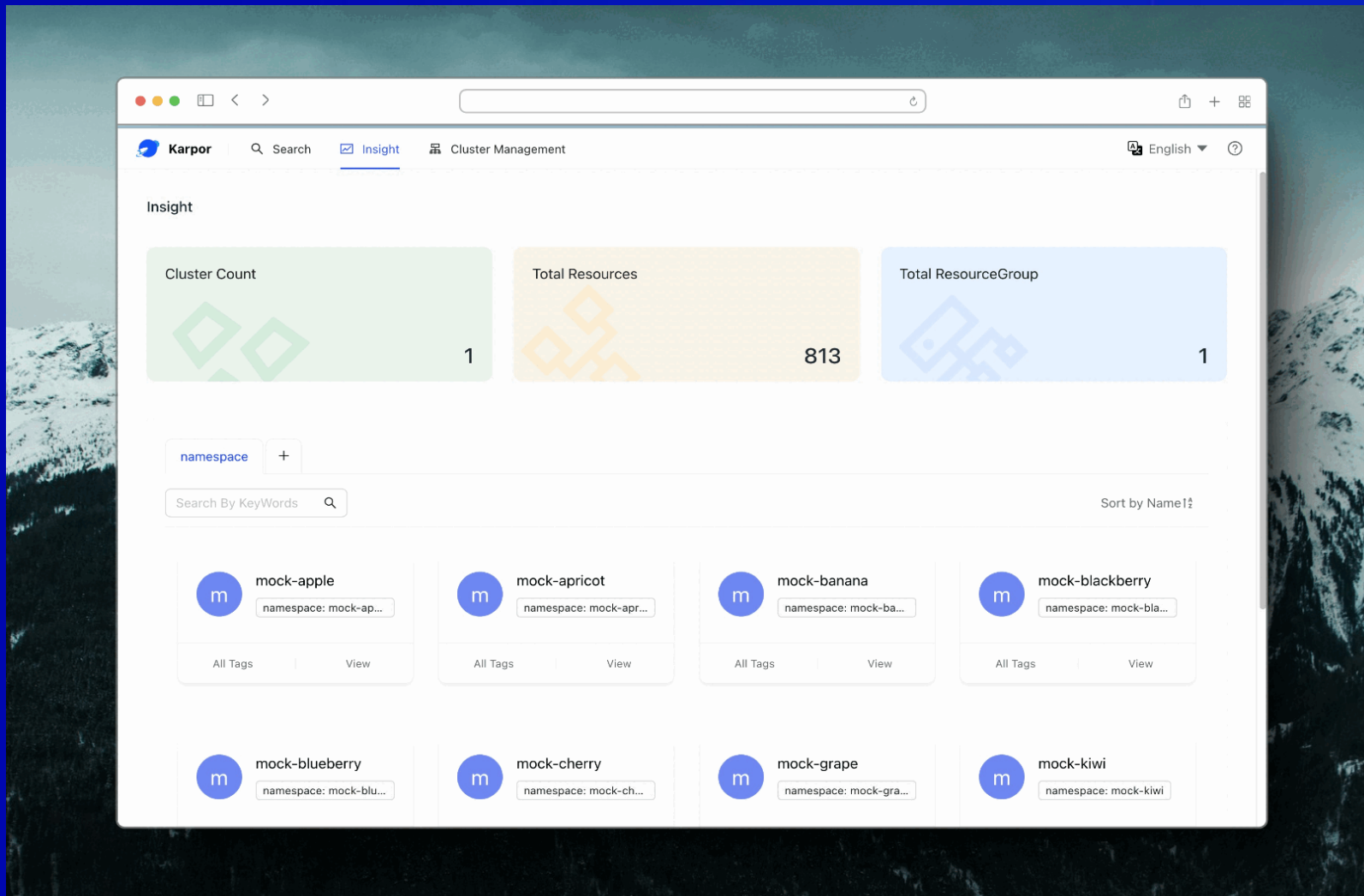
Aggregated Views: Cluster and Namespace

Cluster Health, Resource Topology, Vulnerabilities, Logs and Events, etc.



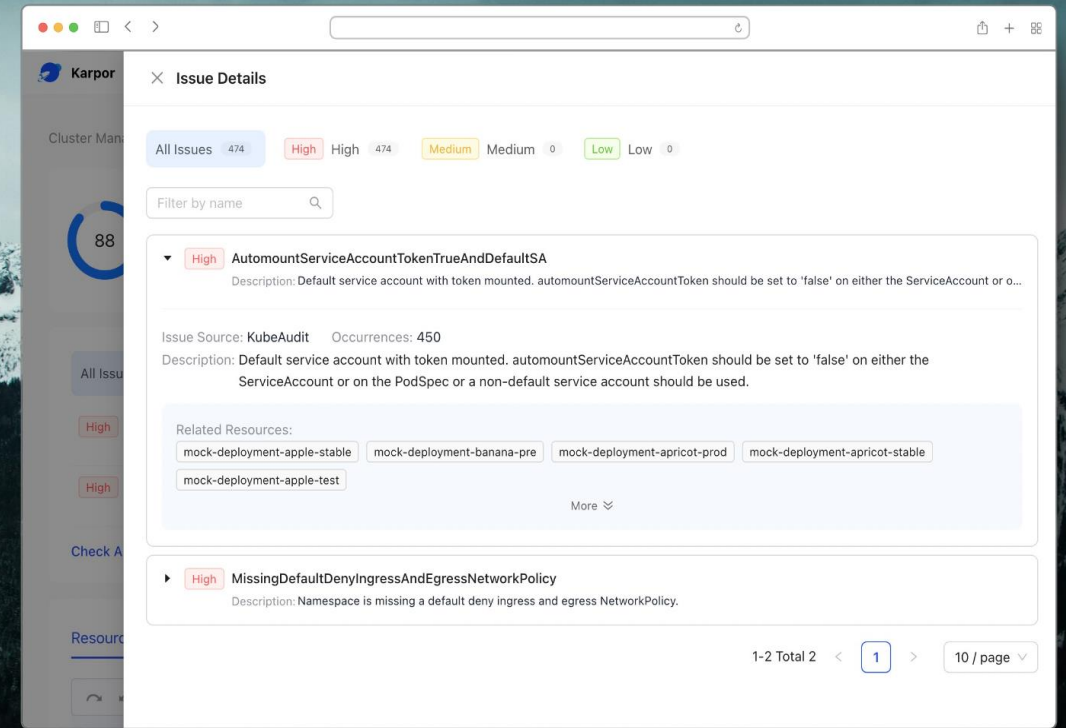
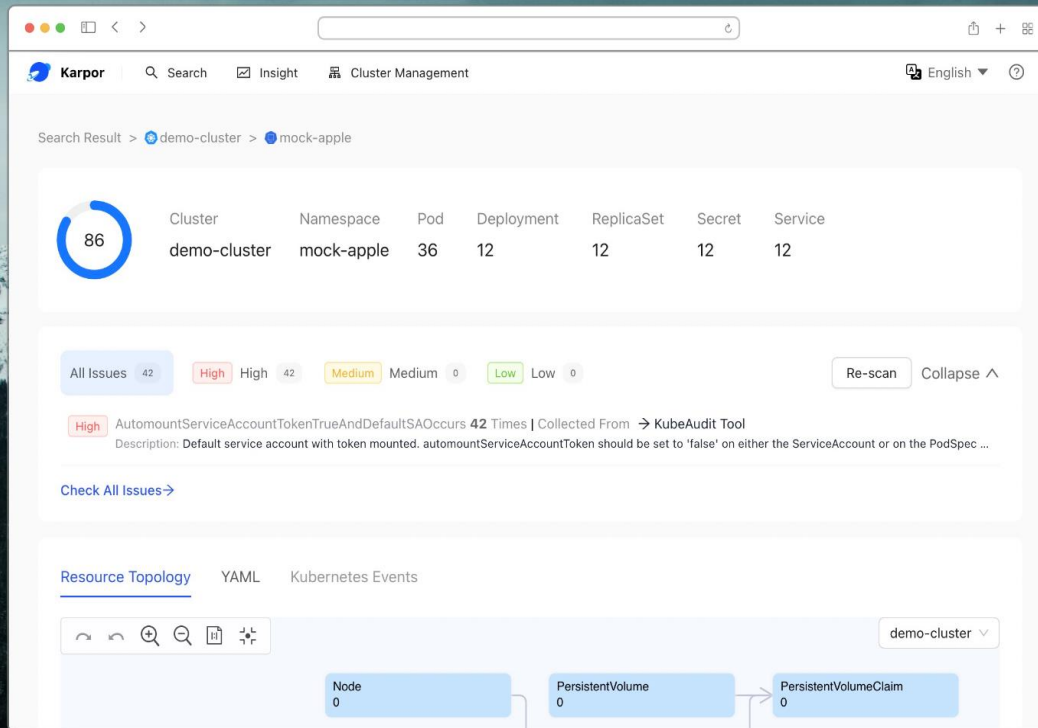
Aggregated Views: Customized Logical Groups

Customized Resource Groups, e.g. application resources cross clusters



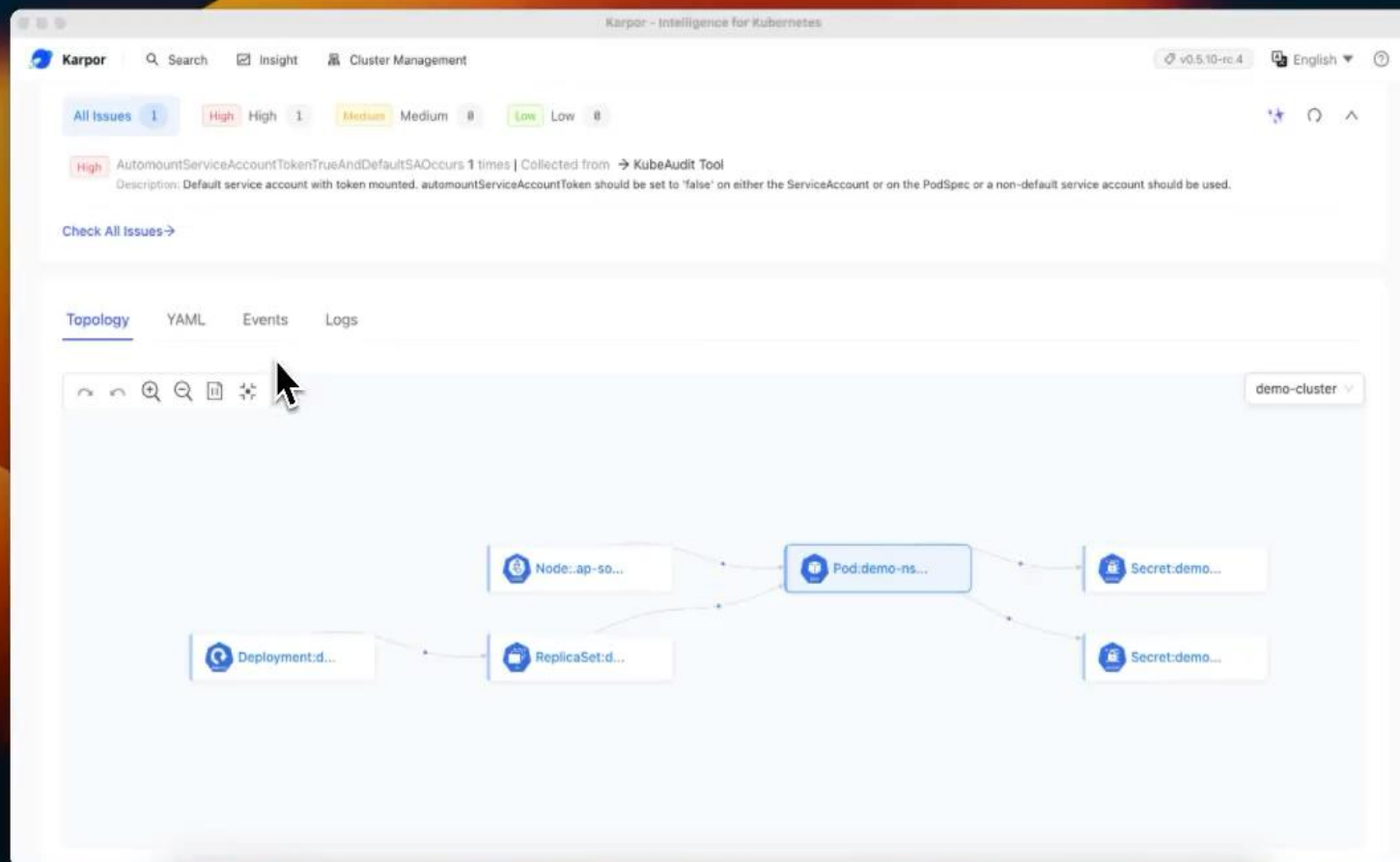
Aggregated Views: Compliance Reports

Analyze and identify security and compliance risks by scanning resources on a regular basis



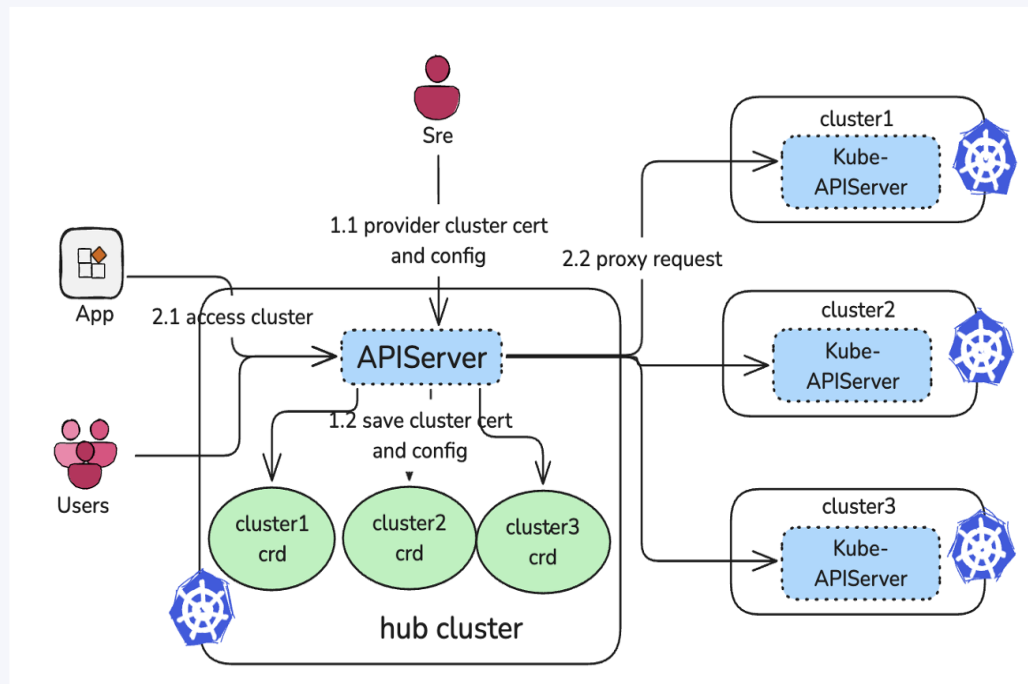
Intelligence for Kubernetes: AI-Empowered Insights

Analyze resource YAMLs, compliance issues, Kubernetes events and logs to excavate deeper insights

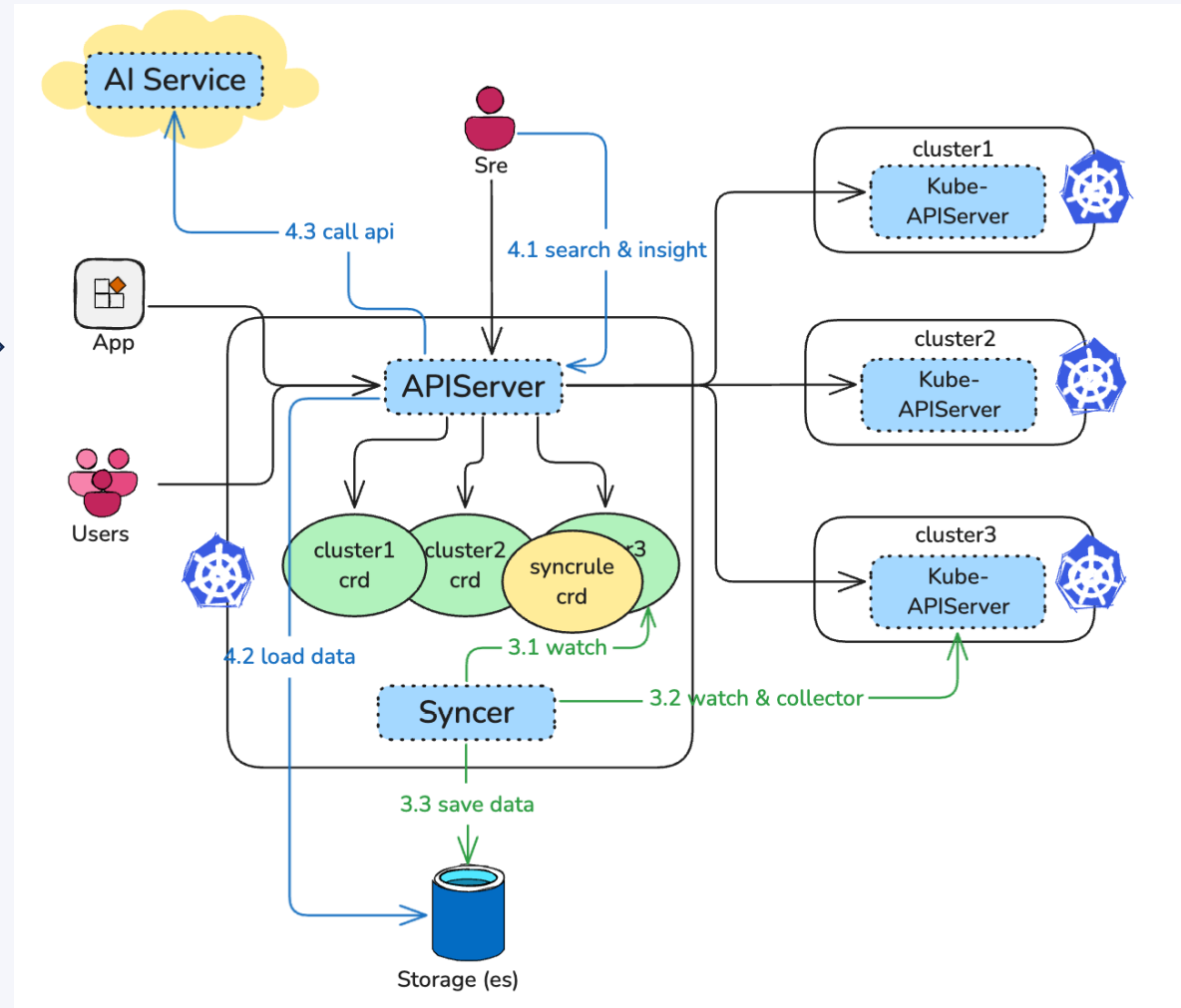


Adoption at Ant Group: From OCM to Karpor

OCM



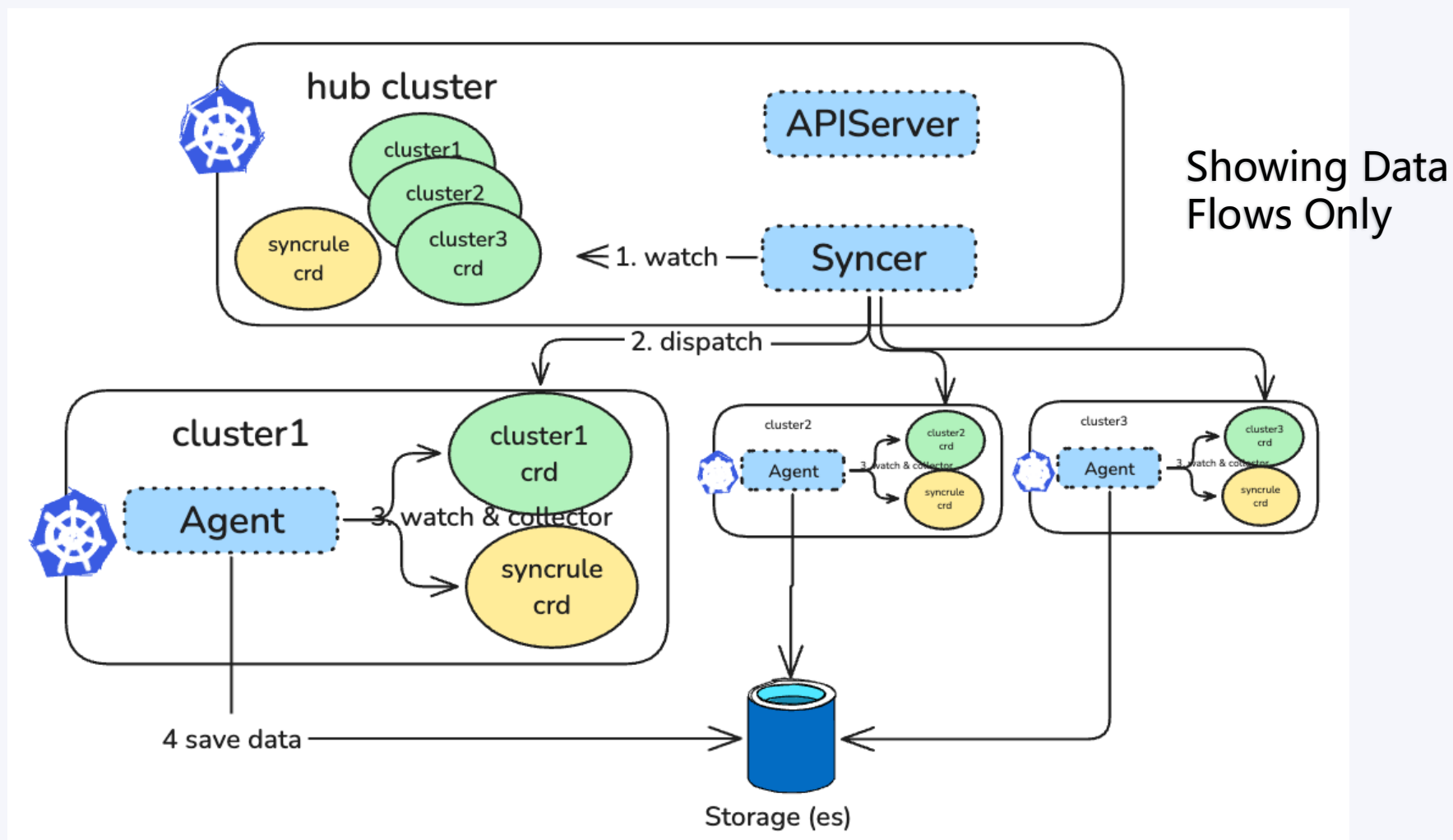
Karpor



Pain points of Syncer ->

- Too many clusters. Syncer becomes a SPOF
- Clusters are huge in size. Syncer consumes a ton of resources

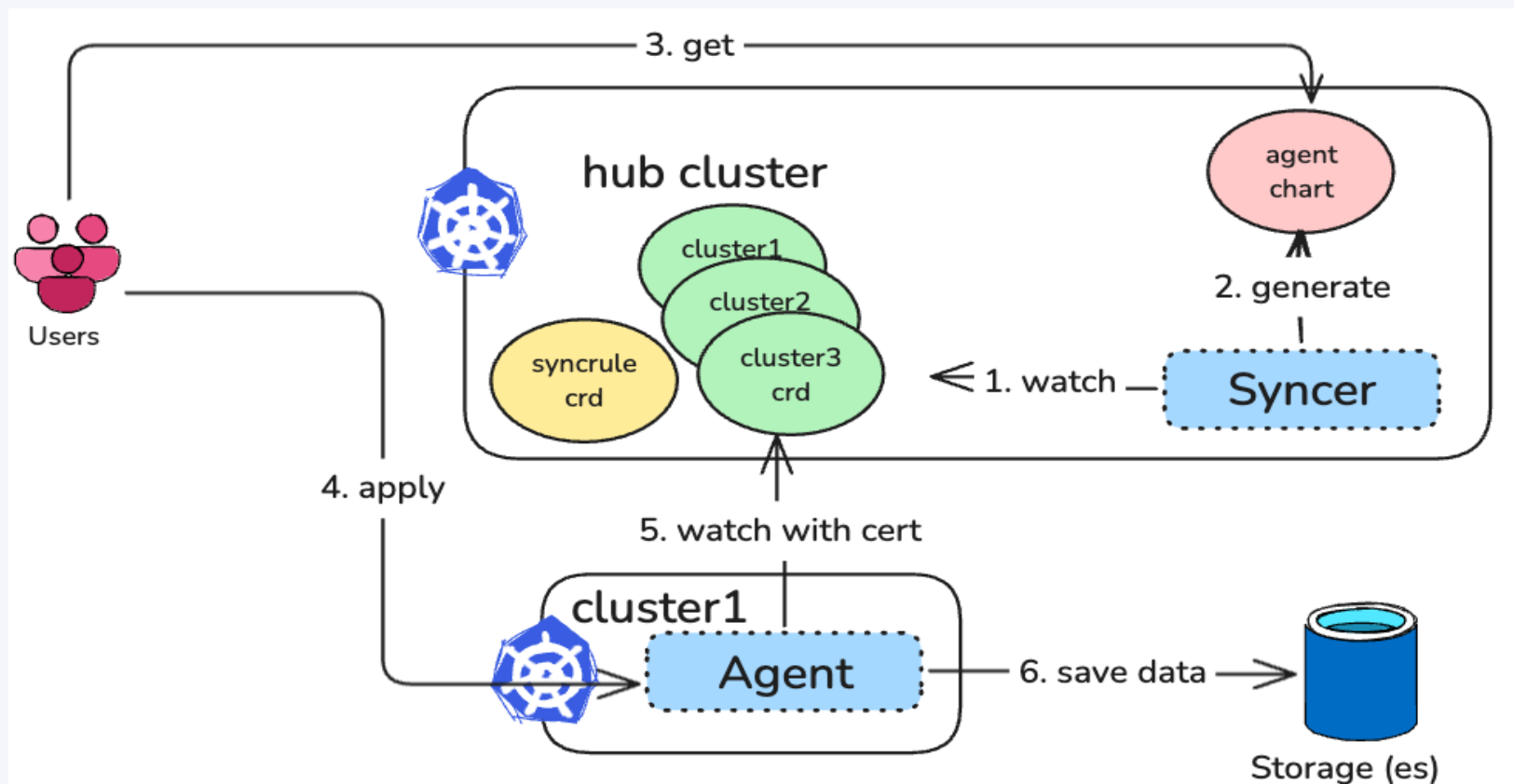
Adoption at Ant Group: Push Mode



Syncer Agent in Spoke Clusters with a Push Mode.

This means Karpenter needs to get write permissions to create resources in Spoke Clusters.

Adoption at Ant Group: Pull Mode



Showing Data
Flows Only

Syncer Agent in Spoke Cluster with a Pull Mode.
No need for Karpenter to get write permissions.

What are we planning?

- 📚 AI Empowerment: Further embrace AI to provide deeper insights - <https://github.com/KusionStack/karpor/issues/634>
- ✈️ Support more clouds: Import Managed K8s on public clouds - <https://github.com/KusionStack/karpor/issues/458>
- 📷 Cluster registration with GitOps and Terraform: <https://github.com/KusionStack/karpor/issues/790>
- 🦋 Scaling: Optimizations in large-cluster adoption - <https://github.com/KusionStack/karpor/issues/787>
- 🍷 Feel free to join us!

The screenshot shows the GitHub repository for KusionStack/karpor. The 'Issues' tab is selected, showing a list of 72 open issues. Three specific issues are highlighted with green boxes at the top: 'Community tasks | 新手任务清单' (Issue #463), 'Wanted: Are you using Karpor | 如果你正在使用 Karpor' (Issue #343), and 'Planning: v0.7.0' (Issue #723). Below these, a search filter 'is:issue state:open' is applied. The main list of issues includes:

- Installation: Add Helm Parameter for Ingress (Issue #808)
- Integration: Integrate a Timeline-based Kubernetes Resource Backtracking Tool (Issue #802)
- Karpor behind ingress getting error recv() failed (104: Connection reset by peer) while reading upstream (Issue #801)
- Feat: Expand the AI Interpret Result (Issue #796)
- Documentation: Cluster Core Concept Documentation (Issue #791)
- Integration: Cluster Registration Terraform Provider for GitOps Integration (Issue #790)
- Integration: Support Deepseek as New AI Backend in Karpor (Issue #790)

Follow us for the latest events!

- Website
 - <https://kusionstack.io>
- Github
 - <https://github.com/KusionStack>
- Slack
 - <https://kusionstack.slack.com>
- Medium
 - <https://blog.kusionstack.io>
- Twitter
 - [@KusionStack](https://twitter.com/KusionStack)



WeChat Group Invitation



Ding Group Chat

KusionStack 官方用户群
111人



扫一扫群二维码，立刻加入该群。



Thanks for listening!

THANKS

2025.08.19

